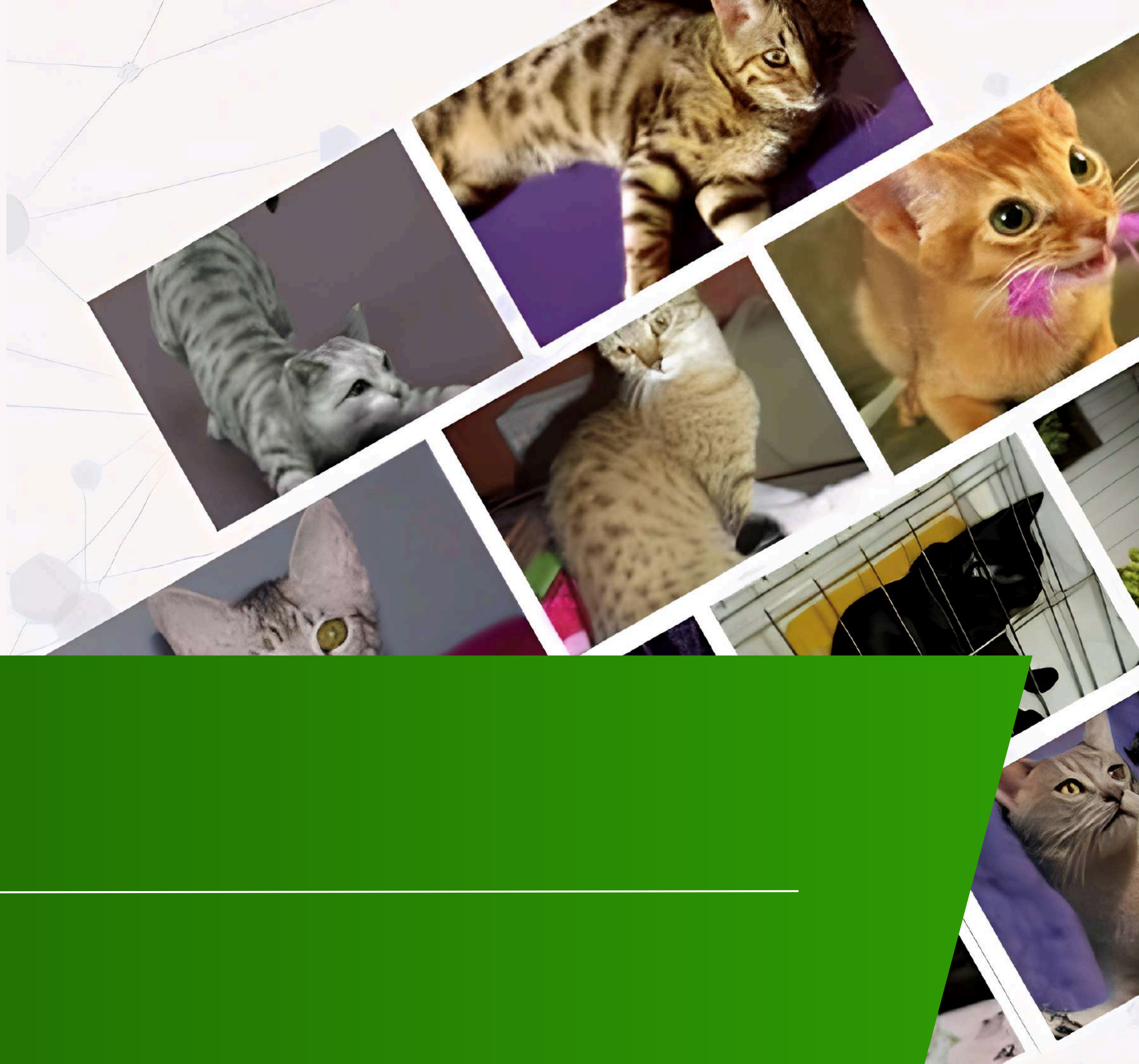


CSE428

IMAGE

PROCESSING

Md. Imam Hasan
Reshad Ul Karim
Syeda Maliha Tabassum



DATASET

THE OXFORD-IIIT PET DATASET

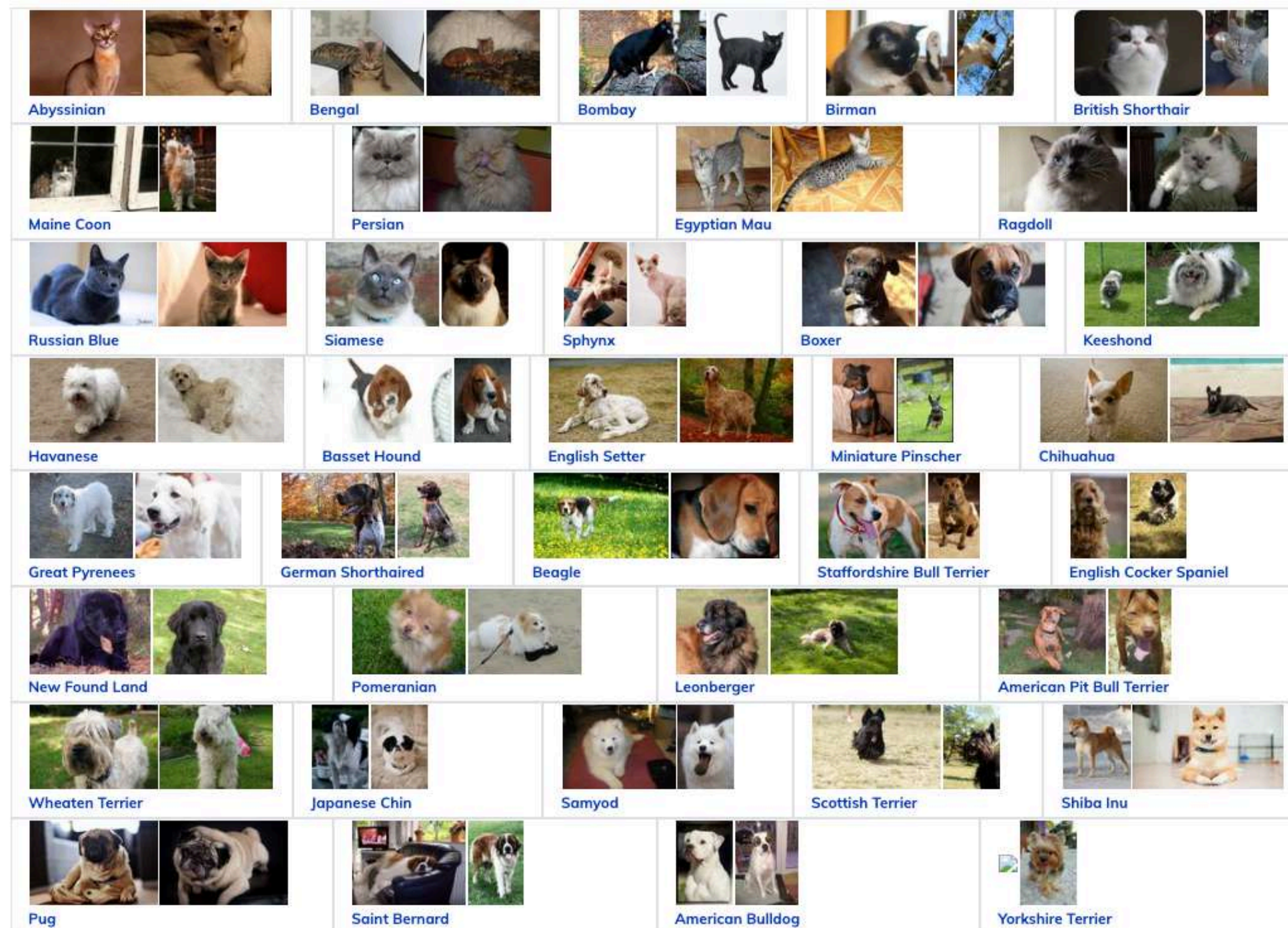
Omkar M Parkhi and Andrea Vedaldi and Andrew Zisserman and C. V. Jawahar

Overview

a 37 category pet dataset with roughly 200 images for each class. The images have a large variations in scale, pose and lighting. All images have an associated ground truth annotation of breed, head ROI, and pixel level trimap segmentation.

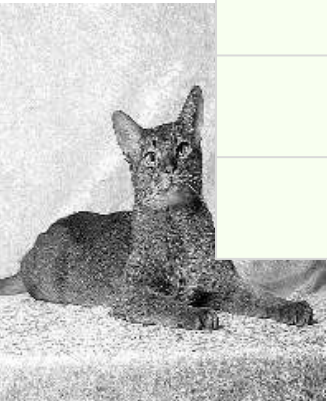
Dataset Statistics

Breed	Count	Breed	Count
American Bulldog	200	Abyssinian	198
American Pit Bull Terrier	200	Bengal	200
Basset Hound	200	Birman	200
Beagle	200	Bombay	200
Boxer	199	British Shorthair	184
Chihuahua	200	Egyptian Mau	200
English Cocker Spaniel	196	Main Coon	190
English Setter	200	Persian	200
German Shorthaired	200	Ragdoll	200
Great Pyrenees	200	Russian Blue	200
Havanese	200	Siamese	199
Japanese Chin	200	Sphynx	200
Keeshond	199	Total	2371
Leonberger	200	2.Cat Breeds	
Miniature Pinscher	200	Family	Count
Newfoundland	196	Cat	2371
Pomeranian	200	Dog	4978
Pug	200	Total	7349
Saint Bernard	200	3.Total Pets	
Samoyed	200	1.Dog Breeds	
Scottish Terrier	199	Breed	Count
Shiba Inu	200	American Bulldog	200
Staffordshire Bull Terrier	189	American Pit Bull Terrier	200
Wheaten Terrier	200	Basset Hound	200
Yorkshire Terrier	200	Beagle	200
Total	4978	Boxer	199

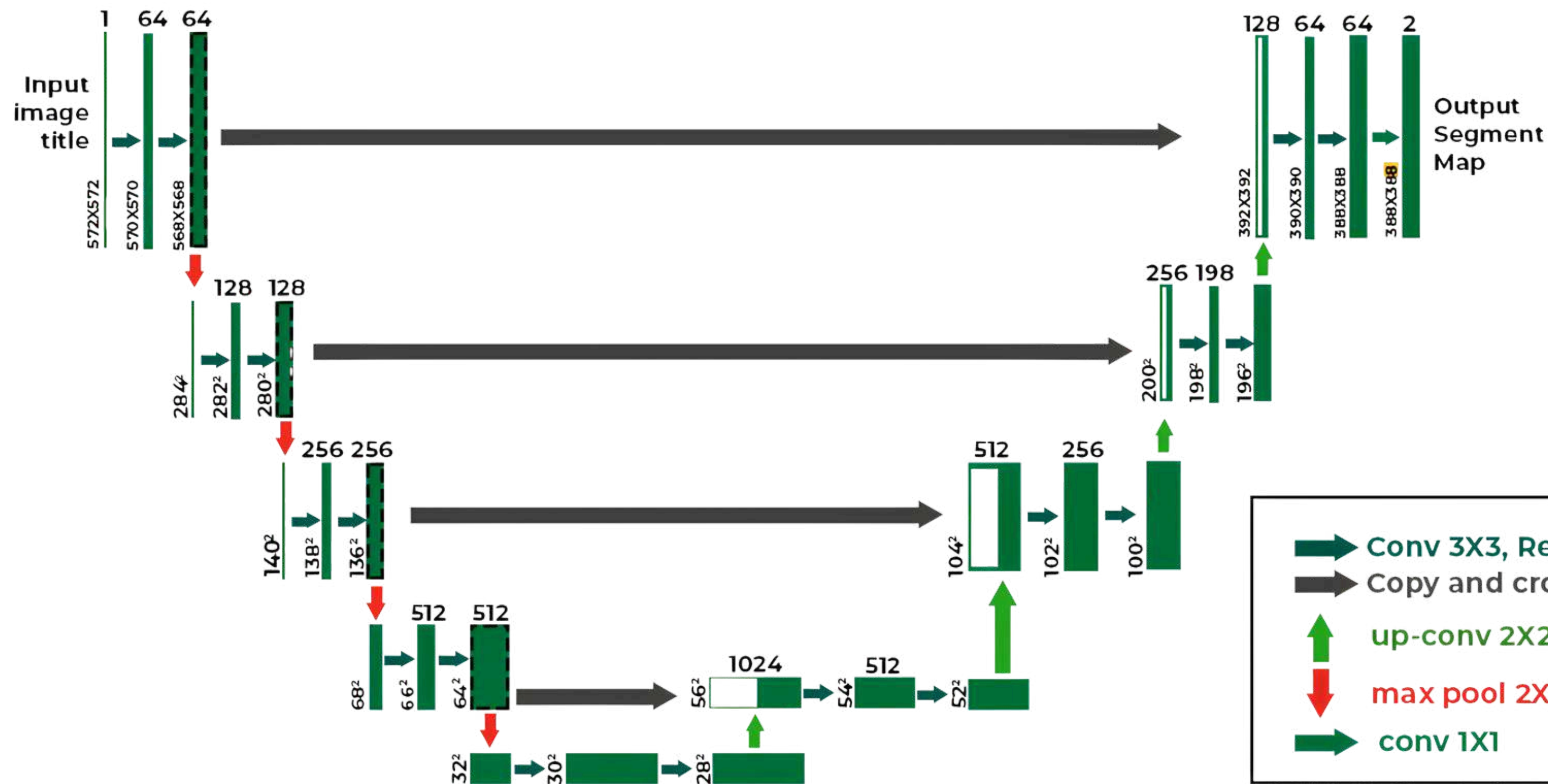


AUGMENTATION & PREPROCESSING

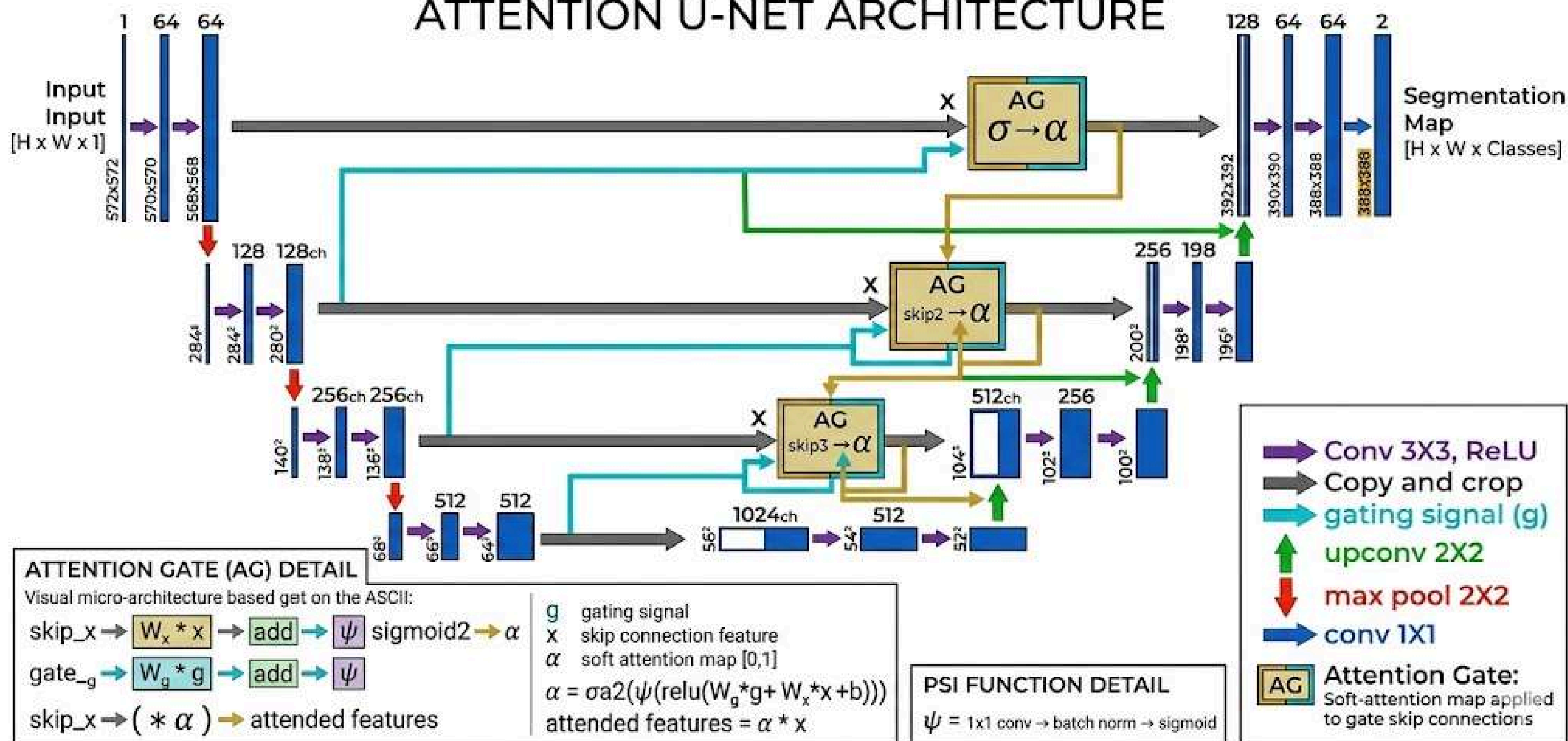
Stage	Transform / Operation	Applied To	Purpose
Input	Image loaded as HWC uint8 NumPy array	Image	Uses OpenCV/dataset-loaded image directly without PIL conversion
Input	Mask loaded as 2D NumPy array	Mask	Provides pixel-level segmentation label map
Geometric Augmentation	Horizontal Flip, p=0.5	Image + Mask	Increases left-right variation and improves generalisation
Geometric Augmentation	Rotation, limit $\pm 15^\circ$, p=0.5	Image + Mask	Makes the model robust to small orientation changes
Geometric Augmentation	Random Resized Crop, size 256×256, scale 0.75–1.0	Image + Mask	Simulates object scale and crop variation
Photometric Augmentation	Color Jitter, p=0.6	Image only	Changes brightness, contrast, saturation, and hue to improve robustness to lighting variation
Photometric Augmentation	Grayscale Conversion, p=0.1	Image only	Encourages the model to learn shape and texture instead of relying only on colour
Noise / Blur Augmentation	Gaussian Blur, p=0.2	Image only	Simulates slight camera blur and improves robustness
Noise / Blur Augmentation	Gaussian Noise, p=0.2	Image only	Makes the model more tolerant to sensor/image noise
Regularisation	Coarse Dropout, p=0.4	Image only	Randomly removes image patches to reduce overfitting and force partial-view learning
Normalisation	ImageNet Mean and Standard Deviation	Image only	Standardises image values for stable model training
Tensor Conversion	ToTensorV2()	Image + Mask	Converts NumPy arrays into PyTorch tensors
Output	Image tensor shape: [3, 256, 256]	Image	Ready for CNN/UNet-style model input
Output	Mask tensor shape: [256, 256]	Mask	Ready for segmentation loss calculation



U-NET

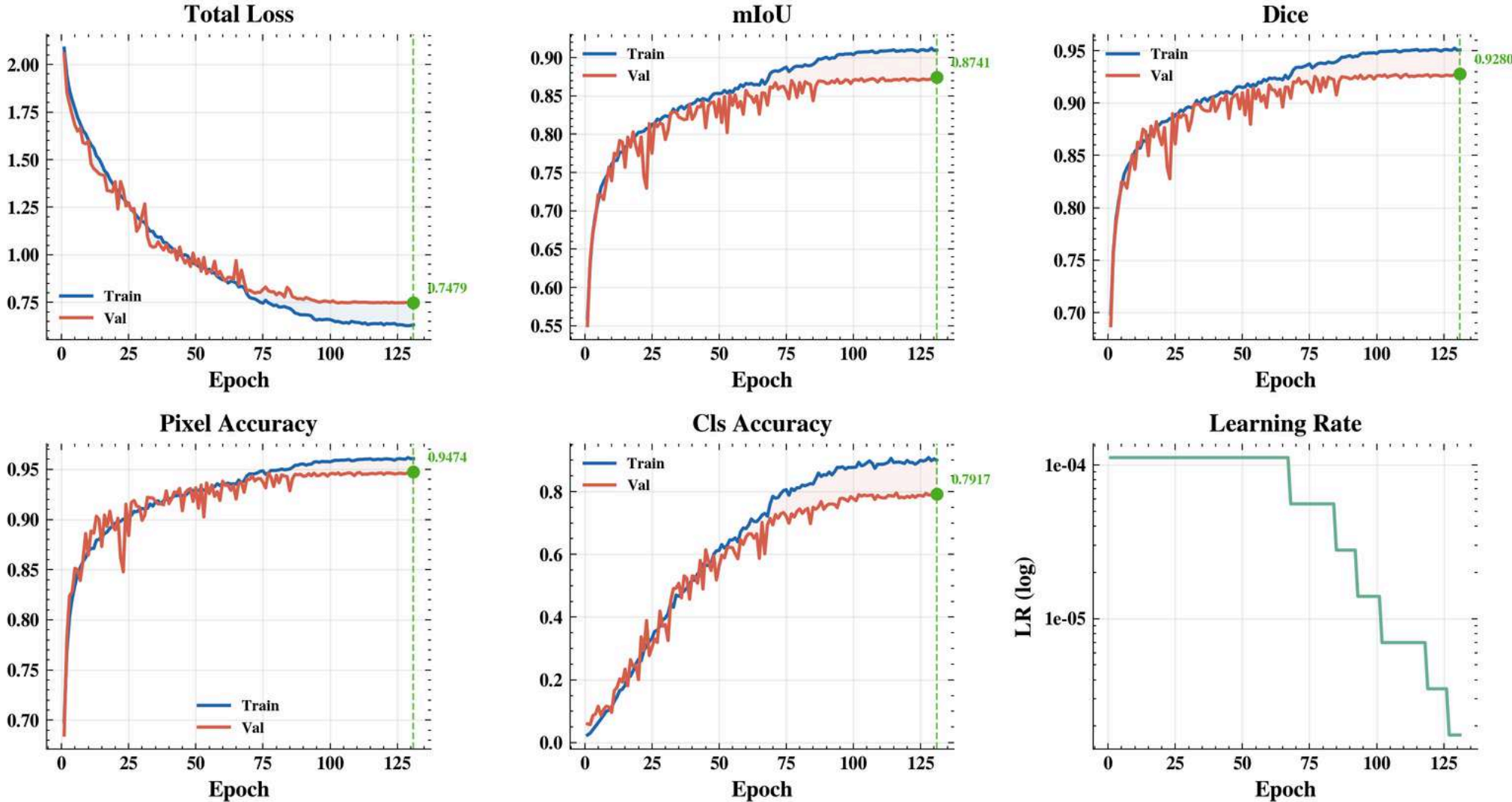


ATTENTION U-NET ARCHITECTURE

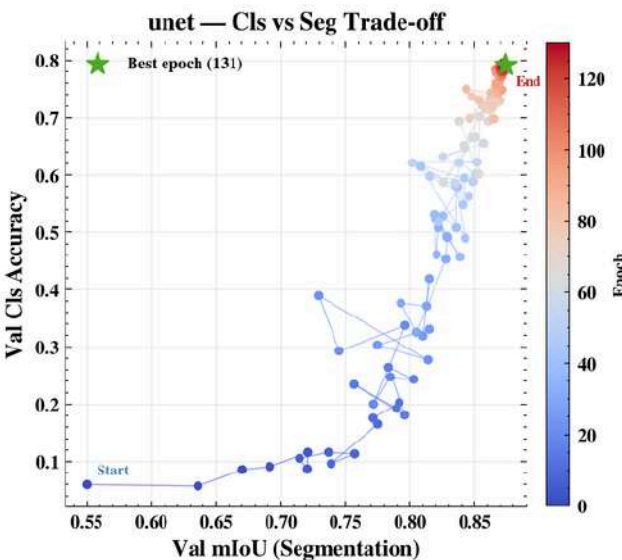
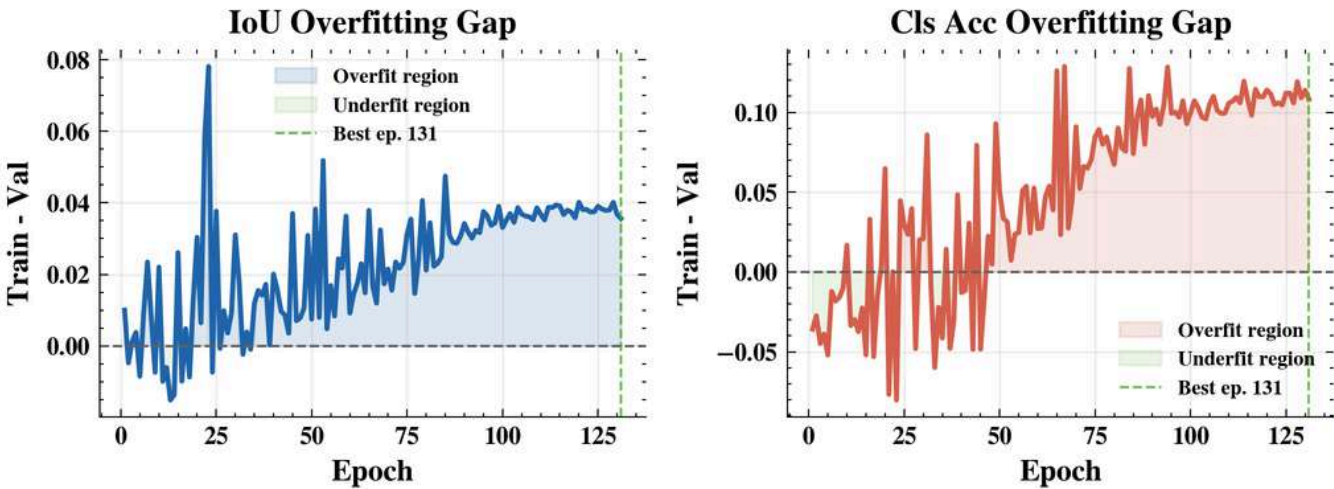


UNET

UNET — Training Dashboard



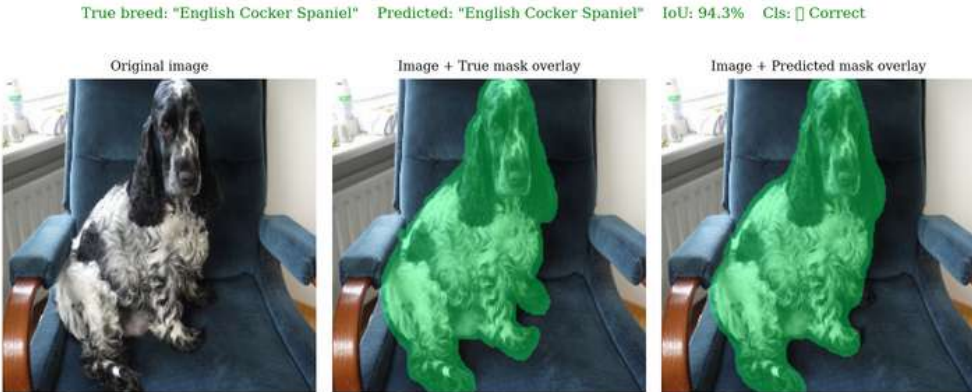
UNET - Overfitting Diagnosis



HYPERPARAMETER TUNED

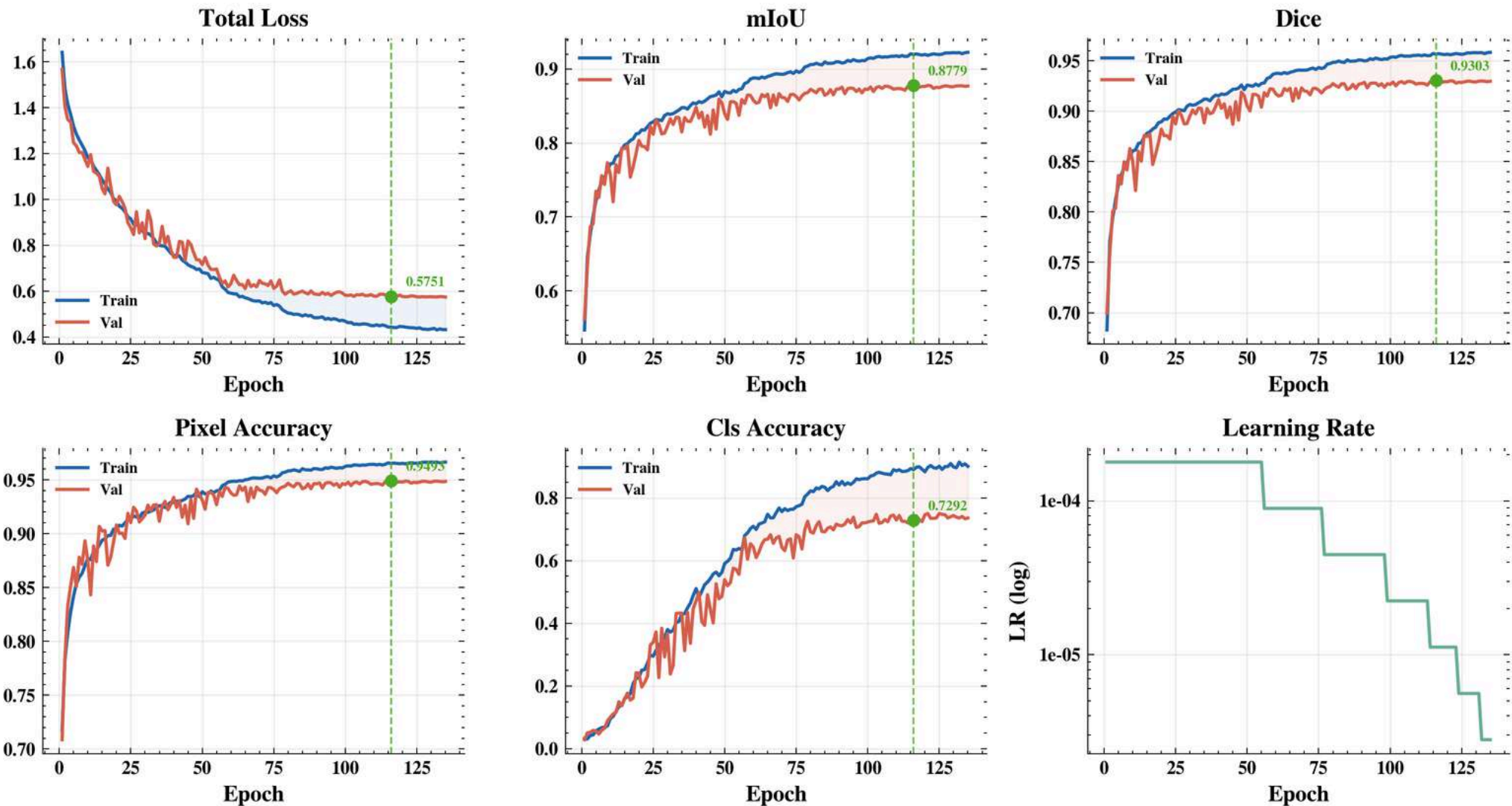
Category	Metric / Parameter	Value
Optimization Summary	Objective Metric	val_iou
	Best Val IoU	0.761673
	Best Trial Number	13
	Total Trials	30
Best Hyperparameters	Epochs per Trial	8
	Learning Rate	0.000117456
	CLS Lambda	0.3599288613
	Weight Decay	0.0004078854
	Dropout Rate	0.5320063444
	CLS WD Multiplier	50
	Scheduler Patience	7

Category	Metric	Value
Segmentation	mIoU	0.884428879591331
	Dice	0.934353059186702
	Pixel Accuracy	0.951269412292621
Classification	Accuracy	0.769692014172799
	Precision	0.776506558544901
	Recall	0.768972762415614
	F-1. Score	0.767437308298722

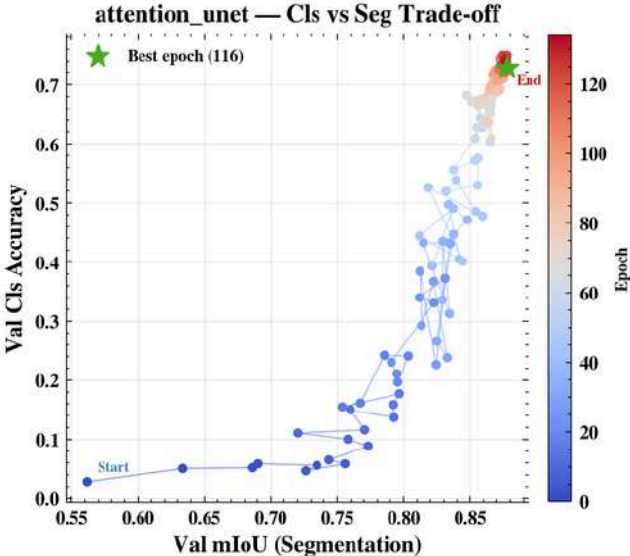
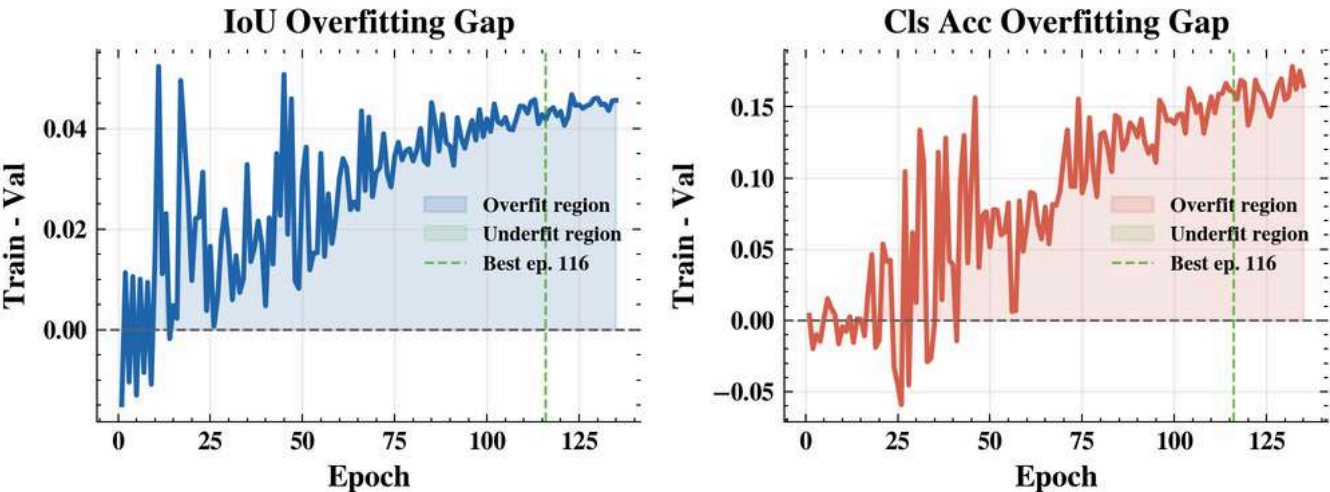


ATTENTION U-NET

attention_unet — Training Dashboard



attention_unet - Overfitting Diagnosis



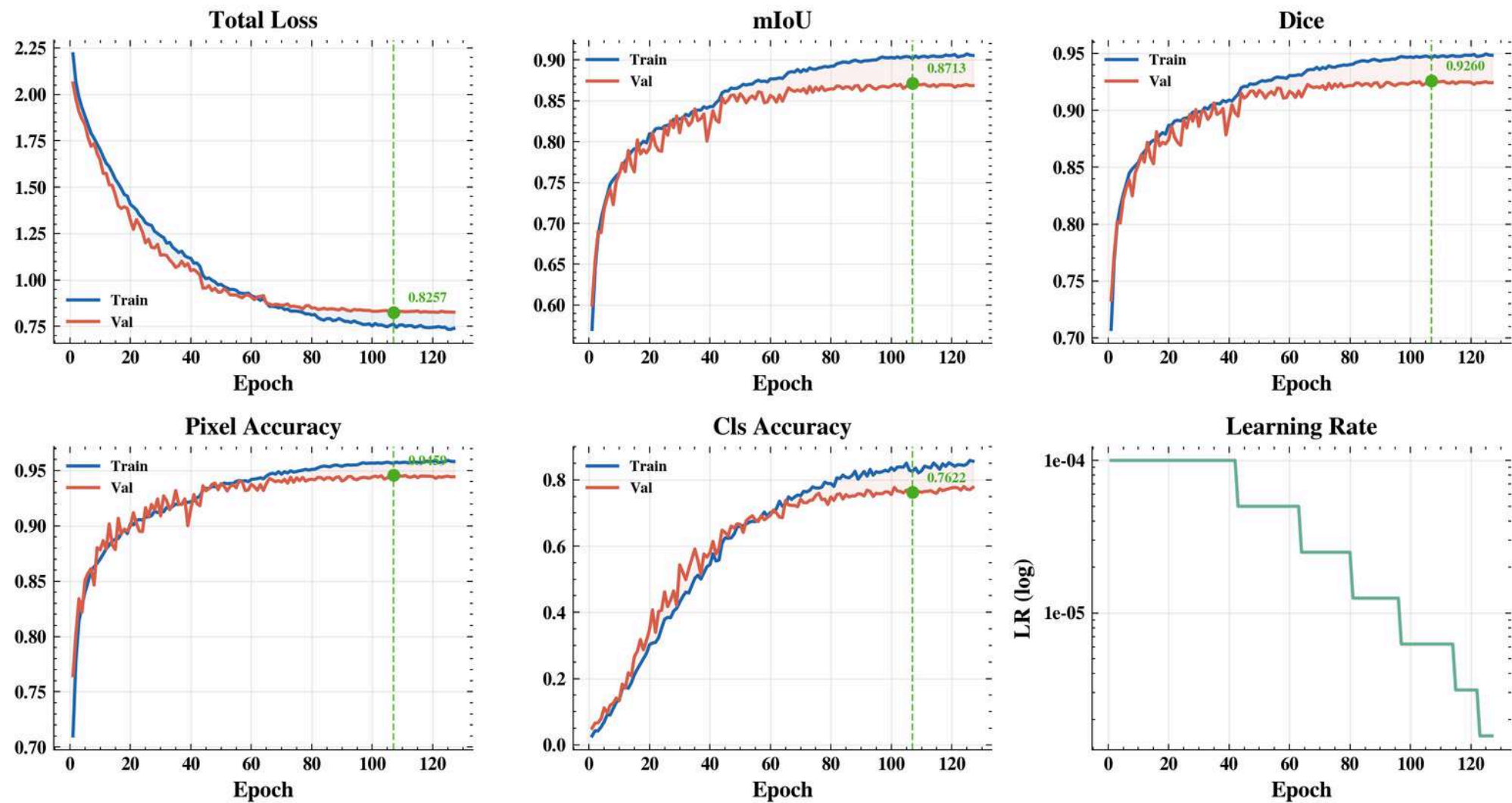
HYPERPARAMETER TUNED

Category	Metric / Parameter	Value
Performance Summary	Objective Metric	val_iou
	Best Validation IoU	0.775684
	Best Trial	23
	Total Trials	30
	Epochs per Trial	8
Optimal Hyperparameters	Learning Rate	0.00017902
	CLS Lambda	0.23606962
	Weight Decay	0.00027469
	Dropout Rate	0.55918269
	CLS WD Multiplier	20
	Scheduler Patience	10

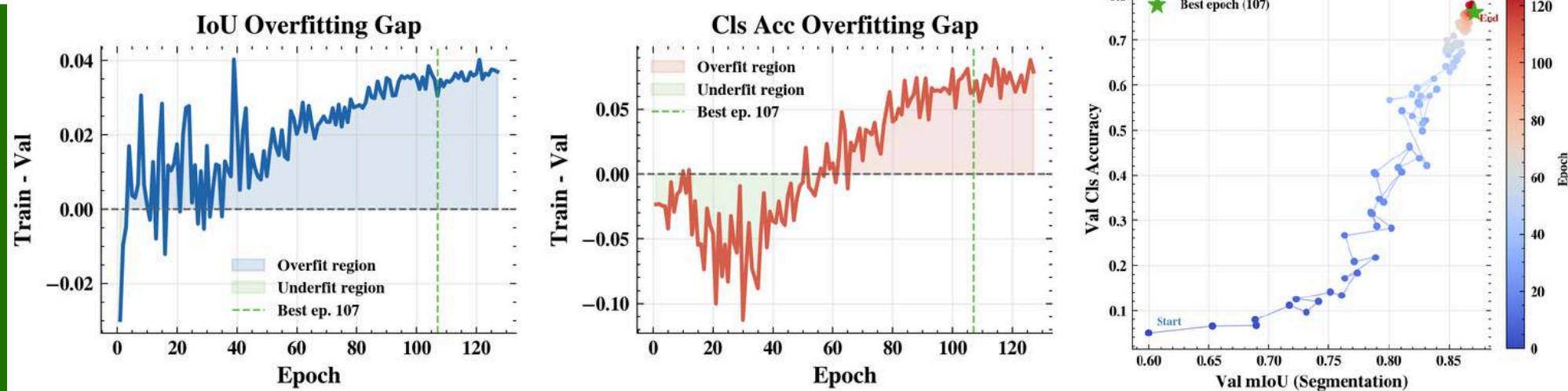
Category	Metric	Value
Segmentation	mIoU	0.889391325719555
	Dice	0.937484473744693
	Pixel Accuracy	0.953580555312546
Classification	Accuracy	0.74897792313982
	Precision	0.758195313244439
	Recall	0.748350567344792
	F1-Score	0.74777114106487

BI-FPN_UNET ADAPTATION FROM EFFICIENTDET

UNET — Training Dashboard



UNET - Overfitting Diagnosis



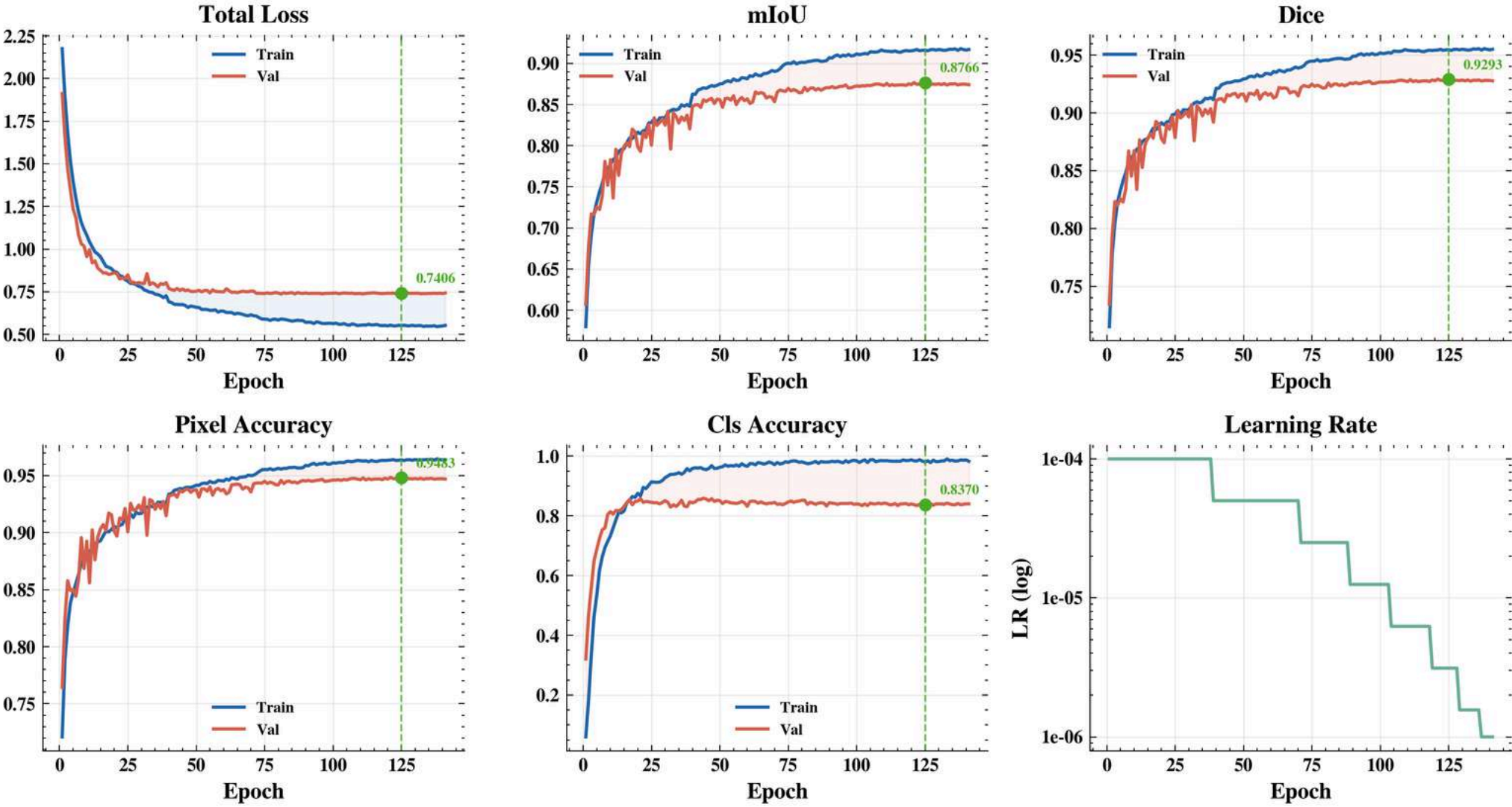
Adapted BiFPN (Bidirectional Feature Pyramid Network) from EfficientDet and integrated in the decoder side of the generic UNET architecture to improve classification accuracy and efficiency.

Category	Metric	Value
Segmentation	mIoU	0.883344389932224
	Dice	0.933676798718732
	Pixel Accuracy	0.950609192063015
Classification	Accuracy	0.749523030798582
	Precision	0.7577678103123
	Recall	0.749050118032345
	F1-Score	0.747535064190547

*hyperparameters are based on empirically tuned unet run

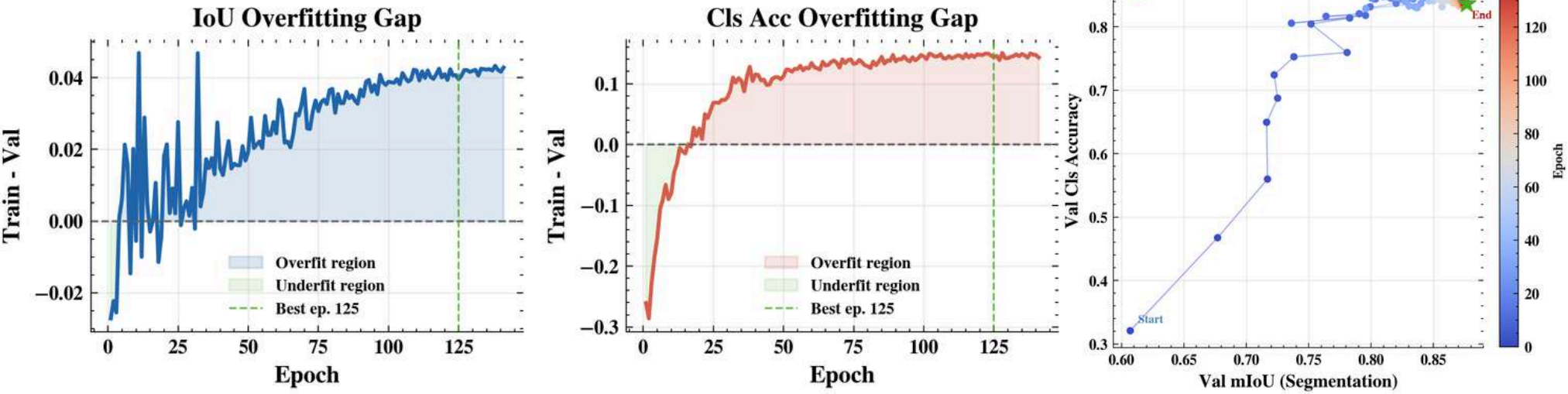
UNET + MOBILENET SMALL CLASSIFIER HEAD

hybrid_mobilenet_v3_small — Training Dashboard



Category	Metric	Value
Segmentation	mIoU	0.889115103154263
	Dice	0.937084490231469
	Pixel Accuracy	0.953413964602961
Classification	Accuracy	0.825565549195966
	Precision	0.82825205767962
	Recall	0.825419086544998
	F1-Score	0.82346807201041

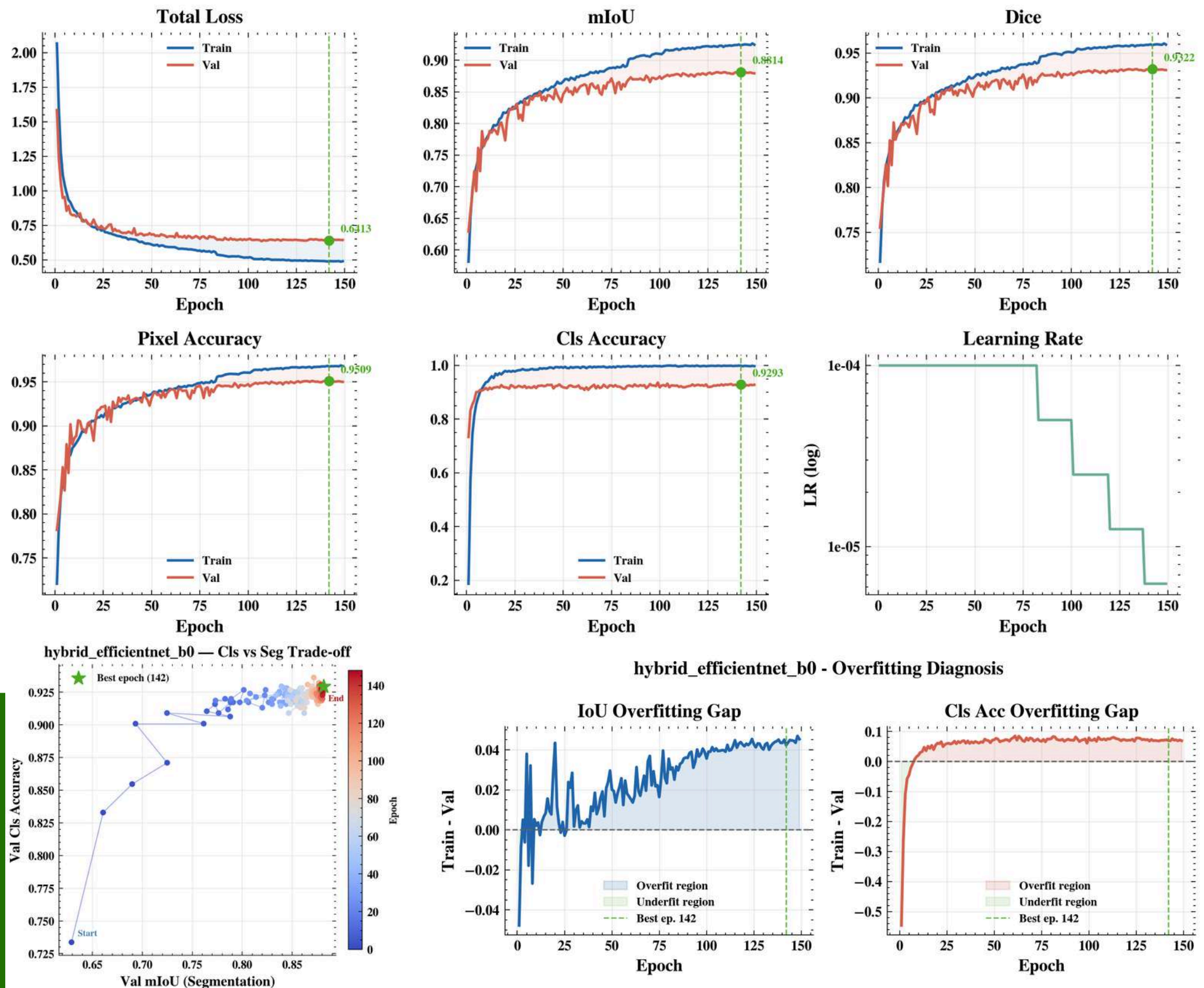
hybrid_mobilenet_v3_small - Overfitting Diagnosis



*hyperparameters are based on empirically tuned unet run

UNET + EFFICIENTNET_B0 CLASSIFIER HEAD

hybrid_efficientnet_b0 — Training Dashboard

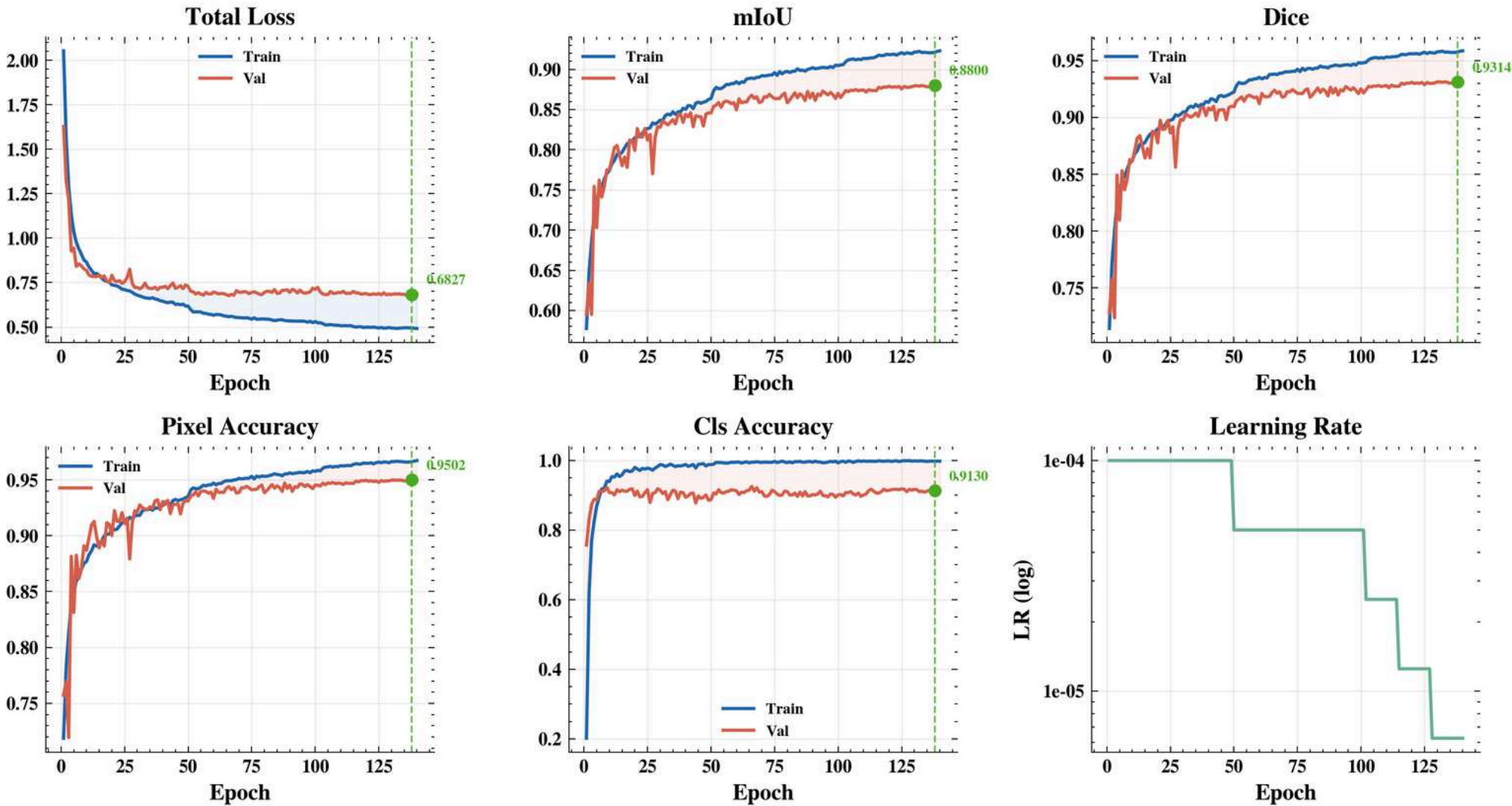


Category	Metric	Value
Segmentation	mIoU	0.89210599961192
	Dice	0.938806806861749
	Pixel Accuracy	0.954810204105358
Classification	Accuracy	0.913872989915508
	Precision	0.914578490814849
	Recall	0.913081913584222
	F1-Score	0.912173020569502

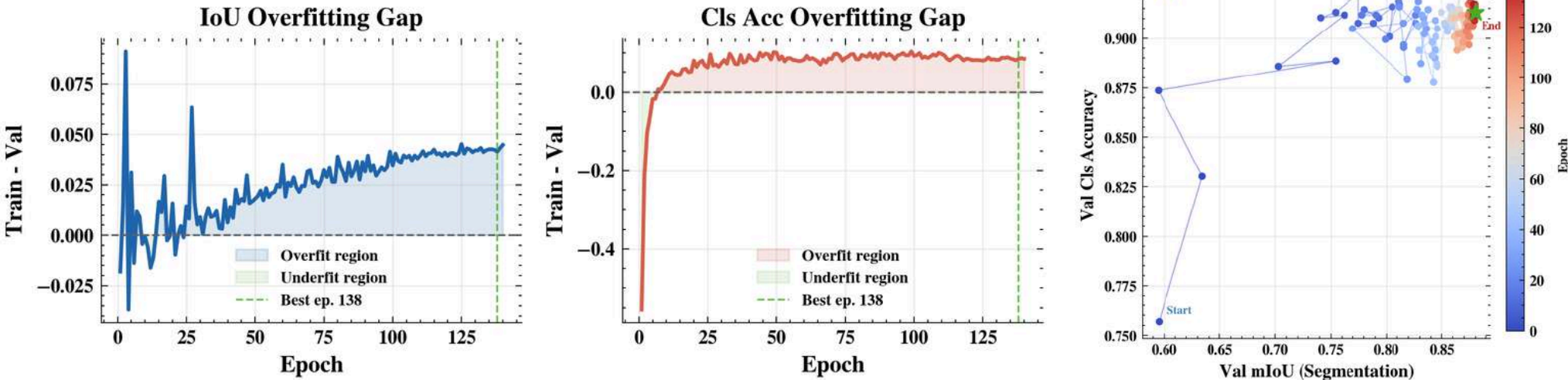
*hyperparameters are based on empirically tuned unet run

UNET + DENSENET CLASSIFIER HEAD

hybrid_densenet121 — Training Dashboard



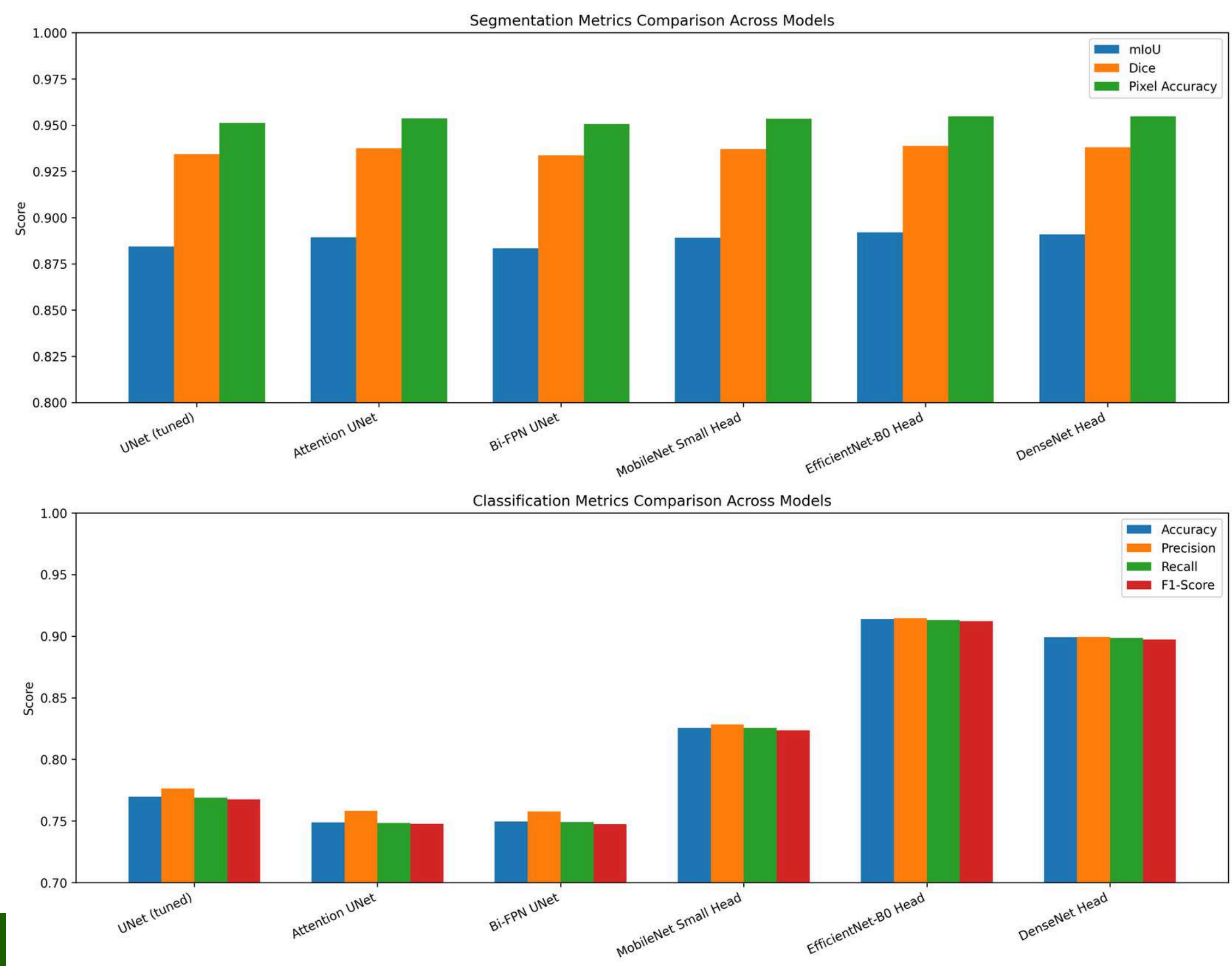
hybrid_densenet121 - Overfitting Diagnosis



Category	Metric	Value
Segmentation	mIoU	0.890984232706614
	Dice	0.937906629672111
	Pixel Accuracy	0.954680518954274
Classification	Accuracy	0.899155083128918
	Precision	0.899466423270173
	Recall	0.898378519526109
	F1-Score	0.897302600129558

*hyperparameters are based on empirically tuned unet run

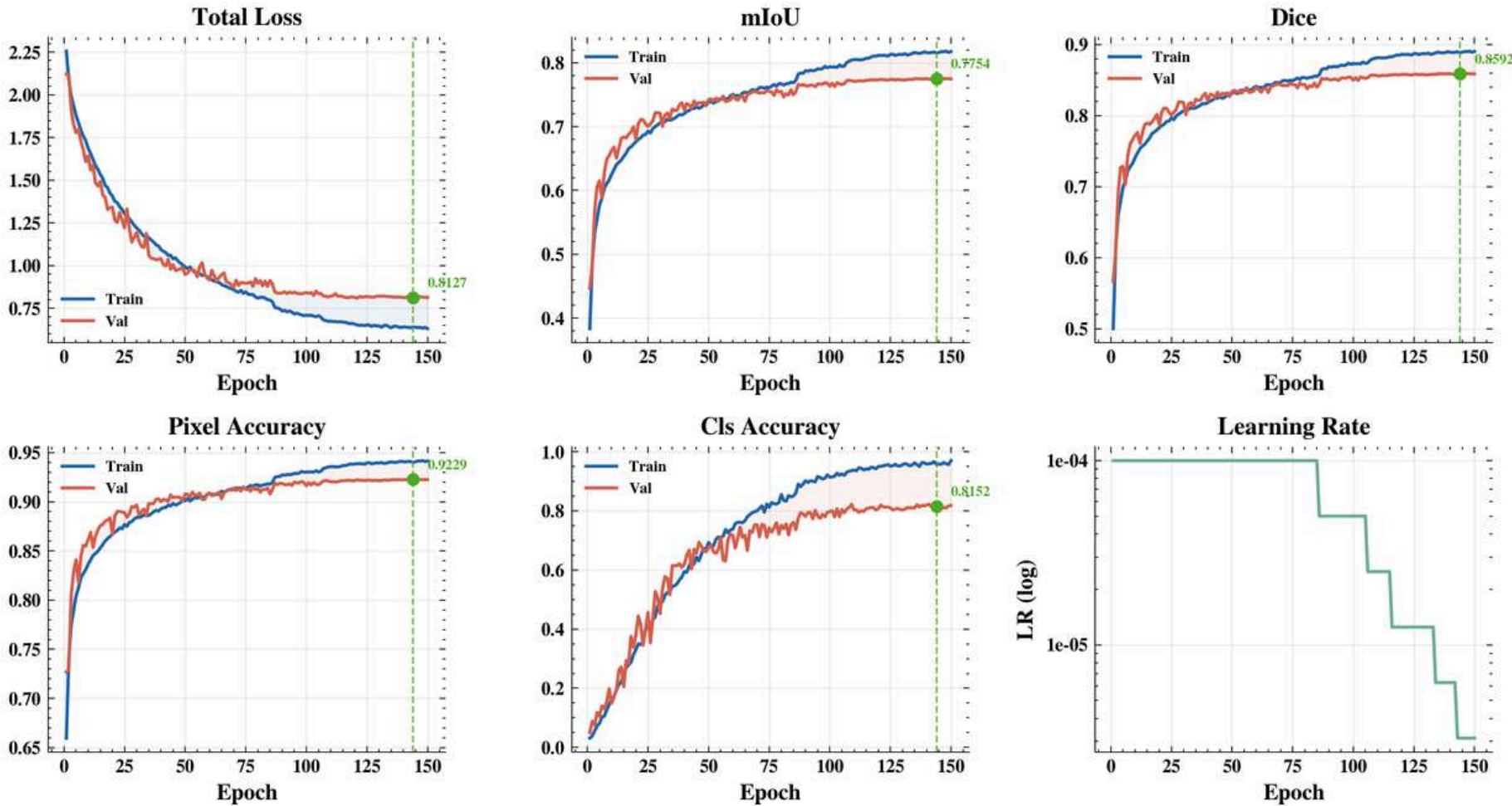
OVERALL COMPARISON



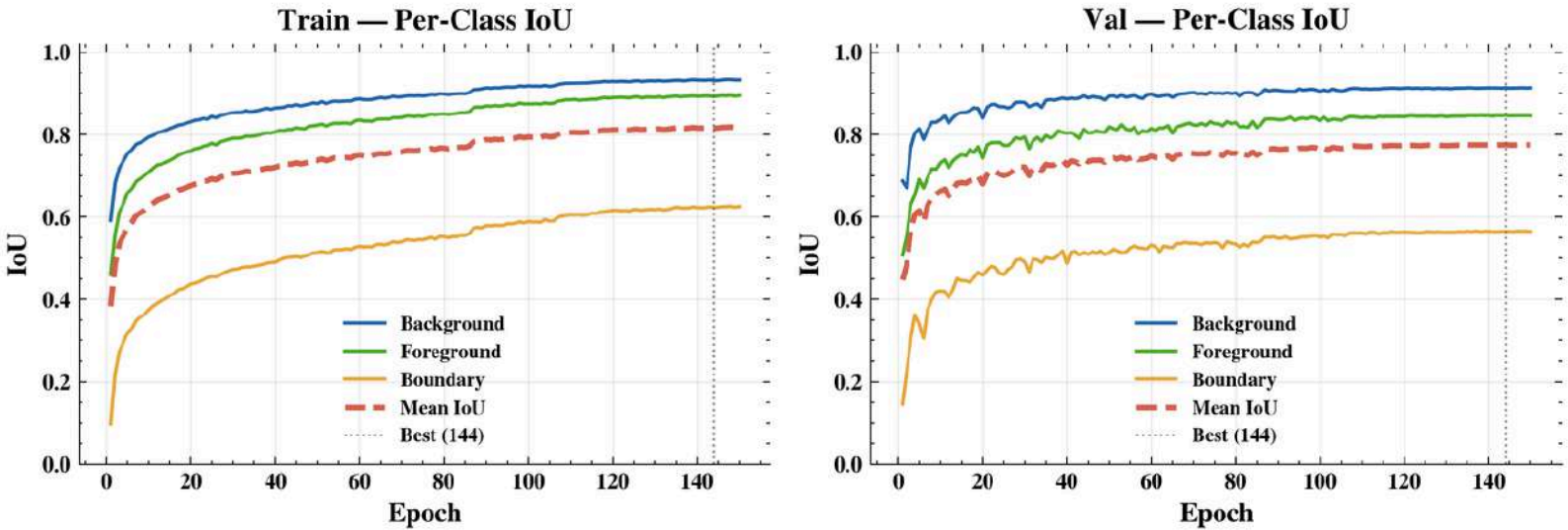
TREATING SEGMENTATION AS A 3 CLASS PROBLEM

- BACKGROUND
- FOREGROUND
- BOUNDARY

UNET_3CLASS — Training Dashboard



UNET_3CLASS — 3-Class Segmentation IoU



Category	Metric	Value
Segmentation	mIoU	0.783981729763141
	IoU (Background)	0.915139409770136
	IoU (Foreground)	0.848055067269698
	IoU (Boundary)	0.58875071224959
	Mean Dice	0.86617309252421
	Pixel Accuracy	0.924982498003089
Classification	Accuracy	0.808394657944944
	Precision	0.811732592897685
	Recall	0.807530711559438
	F1-Score	0.806306884697763

UNET

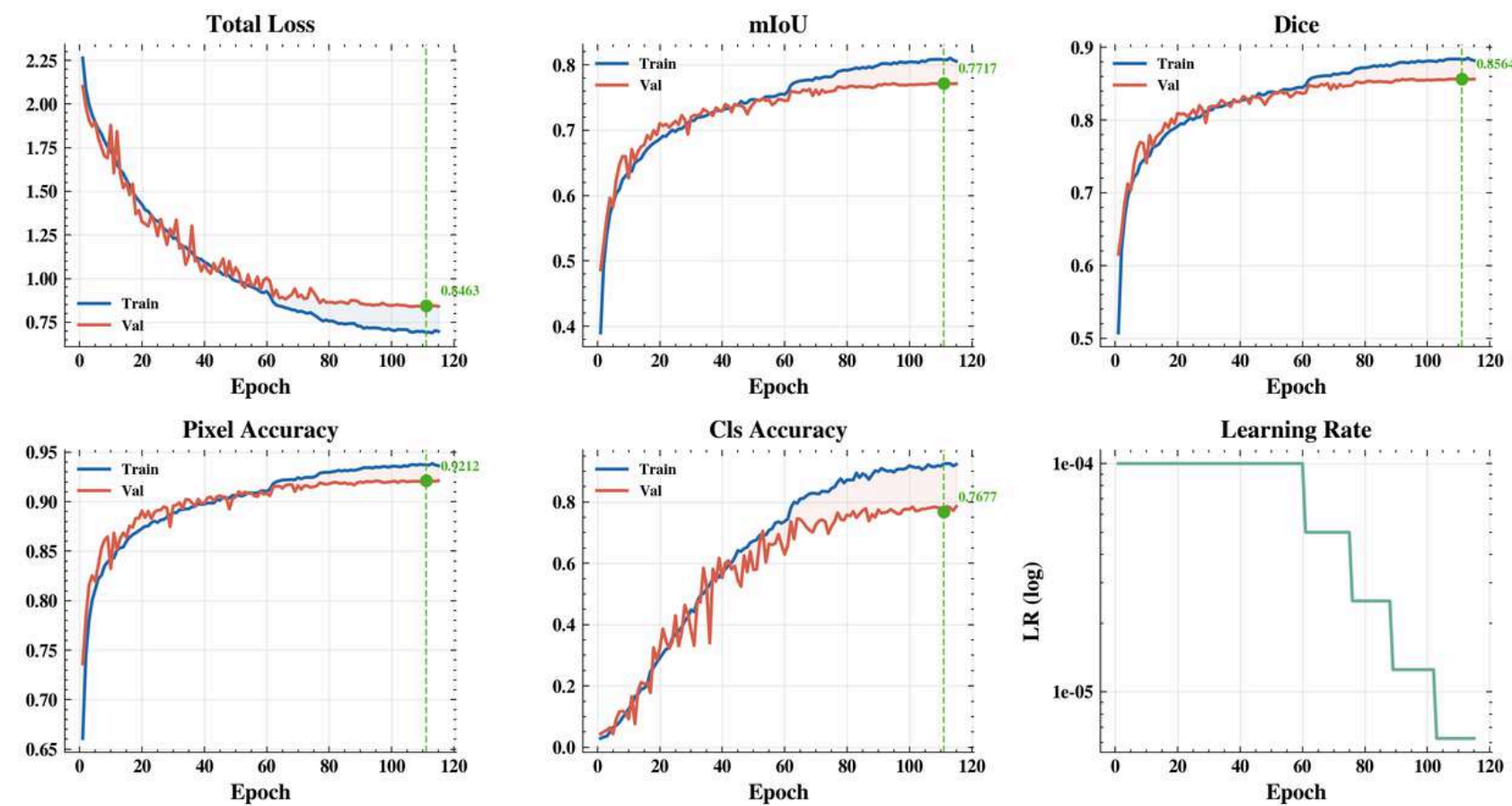
*hyperparameters are based on empirically tuned unet run

TREATING SEGMENTATION AS A 3 CLASS PROBLEM

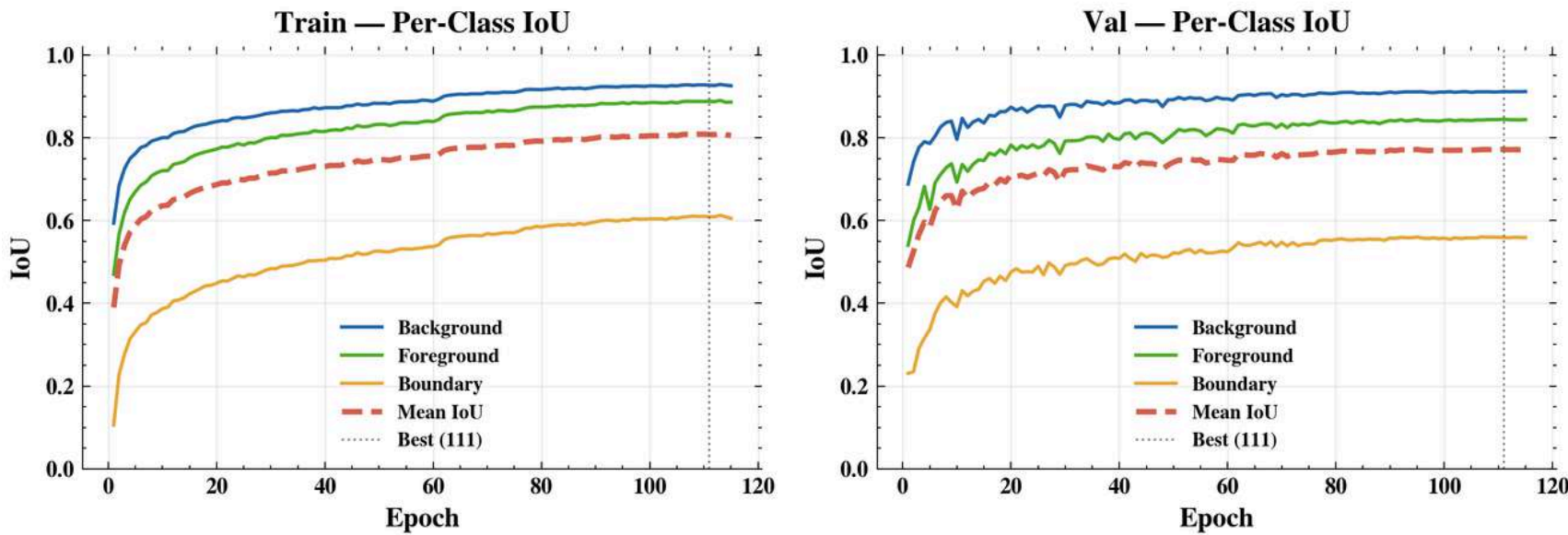
- BACKGROUND
- FOREGROUND
- BOUNDARY

ATTENTION UNET

attention_unet_3class — Training Dashboard



attention_unet_3class — 3-Class Segmentation IoU



Category	Metric	Value
Segmentation	miou	0.77935370120449
	iou_bg	0.91191782173903
	iou_fg	0.84263810696809
	iou_boundary	0.58350517490635
	mean_dice	0.86282533389934
Classification	pixel_acc	0.92265707409900
	accuracy	0.76941946034341
	precision	0.77922150932001
	recall	0.76859063527263
	f1	0.767413134554359

*hyperparameters are based on empirically tuned unet run

INFERENCE SAMPLE

True breed: "Chihuahua" Predicted: "Chihuahua" IoU: 85.1% Cls: ☒ Correct



True breed: "English Cocker Spaniel" Predicted: "English Cocker Spaniel" IoU: 94.3% Cls: ☒ Correct



True breed: "Bengal" Predicted: "Bengal" IoU: 74.8% Cls: ☒ Correct

